

Smooth Flow

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Chapter 1

Introduction

The purpose of this document is to give a comprehensive overview of the scope, modules, use cases, requirements and system design and architecture of our [Final Year Project \(FYP\)](#) titled Smooth Flow. This document is intended to be read by the developers, project managers, users, testers and documentation writers. It mainly serves as a primary resource for understanding project's objectives, scope, design, requirements and implementation.

1.1 Problem Statement

A fluency disorder is an interruption in the flow of speaking characterized by atypical rate, rhythm, and disfluencies (e.g., repetitions of sounds, syllables, words, and phrases; sound [Prolongation](#); and [Blocks](#)), which may also be accompanied by excessive tension, speaking avoidance, struggle behaviors, and secondary mannerisms [[1](#)]

Around 80 million people suffer from speech disfluencies worldwide [[2](#)]. They require in-person speech therapy in order to improve their disfluencies. There are currently no robust digital solutions which cater to the needs of such individuals along with connecting them with Speech Language Pathologist remotely.

1.2 Scope

Our system is a web-based application that is only targeted towards English language speakers who suffer from speech disfluencies. It aims to only assist them in improving their speech fluency through the use of science and evidence backed speech therapy exercises. The correct usage and ability to follow the instructions of the speech therapy exercises is the sole responsibility of the patient. To encourage patients to try the speech exercises daily, there will be [Experience Points \(XP\)](#) awarded and Achievement Badges

for reaching certain milestones in speech therapy exercises. XP allows patient to level up. Levelling system for the patients is a [Gamification](#) element. The amount of XP gained depends on the performance of the speech therapy exercise. Achievement Badges are automatically awarded to the patient when they reach certain milestones. Feedback will be provided to the patient every time they complete a speech therapy exercise. Feedback will inform the patient whether their speech has any more disfluencies or not. Feedback will also include the percentage improvement of the patient's speech based on a severity metric (duration of disfluency and frequency of disfluency [3]). In addition to the speech therapy exercises, patients can consult with an actual [Speech Language Pathologist \(SLP\)](#) registered on the system through text or video.

1.3 Modules

1.3.1 Detection and Multi-Label Classification of Patient's Speech

This module is aimed at the detection and multi-label classification of speech disfluencies in the patient's speech through audio input. Main objective of this module is to assist the patient in identifying the exact nature of their speech disfluencies.

1. The system shall be able to detect the presence of speech disfluency in the recorded audio of the patient.

The patient will be required to read a randomly generated passage. The patient's speech will be recorded by the system. The patient's speech will then be analysed by Machine Learning models for the presence of any speech disfluencies. The results will be displayed on the screen.

2. The system shall be able to classify the audio of patient into multiple speech disfluency classes.
3. The system shall be able to recommend relevant speech therapy exercises based on the classification of the patient's speech.

1.3.2 Speech Therapy Exercises

This module is concerned with the speech therapy exercises that the patient will try in order to improve the fluency of their speech. Each speech therapy exercise follows a gamified approach to better engage the patients and to help with their retention as well.

1. **Reading Passage Exercise:** A passage shall be generated by [Large Language Model \(LLM\)](#). The patient shall read the passage while their speech is being recorded

by the system. The difficulty setting shall determine the number and complexity of the words in the passage. The recorded audio of the patient shall be transcribed into text and compared with the given passage. Any differences detected shall be used for detecting speech disfluencies, measuring severity and measuring performance metric. At the end of the exercise, a feedback report will be displayed which will inform the patient about their performance in the exercise (disfluency severity), how much they improved and difficulty setting of the exercise.

2. **Matching Syllables Exercise:** The patient shall speak challenging words whose syllables would be spaced out. The patient shall input the challenge word themselves in the system. The syllables of the word shall be ticked according to the sound of a [Metronome](#) which shall help the patient with keeping a steady [Beats per Minute \(BPM\)](#) when pronouncing syllables. The patient shall pronounce the given word over 4 different BPMs: 64, 92, 118, 144 with the word repeating 3 times each. The patient's audio shall be recorded and transcribed to text in order to compare their pronounced syllables to the actual word which shall determine their performance. At the end of the exercise, a feedback report will be displayed which will inform the patient about their performance in the exercise (disfluency severity), how much they improved and difficulty setting of the exercise.
3. **Sound Mimic:** A pre-recorded audio shall be played to the user, containing 1-2 clauses that form a sentence. This sentence shall be free of any stuttering. The patient shall then repeat the sentence from the played audio. The speech of the patient would be recorded and transcribed and then compared with the original audio that was played. The difficulty for this exercise shall be increased by using sentences that have more challenging words to pronounce. Any differences detected shall be used for detecting speech disfluencies, measuring severity and measuring performance metric. At the end of the exercise, a feedback report will be displayed which will inform the patient about their performance in the exercise (disfluency severity), how much they improved and the difficulty setting chosen for the exercise.
4. **Breathing Control:** The patient shall follow animations for breathing in and breathing out to help control their anxiousness, promote relaxation and reducing frequency of stuttering moments. The patient shall then read a sentence after the breathing exercises to gauge their performance and impact of the breathing activity on the patient. This activity does not have difficulty options. At the end of the exercise, a feedback report will be displayed which will inform the patient about their performance in the exercise (disfluency severity) and how much they improved.

1.3.3 Baseline of the Patient

This module is concerned with establishing a baseline of the patient's speech. The main objective of this module is to create a user profile for the patient along with estimating the normal rate of speech of the patient. This baseline would assist in the analysis of the patient's speech and checking for any improvement in the fluency.

1. The system shall be able to take input of the patient's details and create a profile for the patient.

An input form will be displayed which the patient will fill according to their specific details. Once confirmed, the system will save the patient's profile.

2. The system shall be able to establish the normal rate of speech of the patient.

This will be done by providing the patient with passages to read from. The difficulty level of the passages will start from high complexity and gradually move towards easy and simple passages. Each time the patient reads the passage, the patient's speech will be checked for any disfluencies. As soon as the analysis of patient's speech shows that there are no more speech disfluencies, the system would then calculate the normal rate of speech of the patient.

1.3.4 Progress Tracking and Feedback Reports

This module is concerned with providing the user with feedback of their speech therapy exercises and allowing the user to track their overall progress of improving their speech.

1. The system shall provide the patients with feedback reports after every speech therapy exercise.

The feedback report provided by the system will contain the patient's performance (disfluency severity) in the speech therapy exercise, difficulty setting of the exercise, how much the patient improved, time taken to complete the exercise, any achievement badges unlocked and the amount of XP gained.

2. The patient will be able to check their history of feedbacks.

All feedbacks will be stored and the patient will be able to view their past feedbacks. This will allow the patient to track their overall progress.

3. The patient will be able to export their feedback reports in PDF format.

1.3.5 SLP Consultation

This module is concerned with connecting the Speech Language Pathologists (SLPs) with the patient. This is to ensure that the patient can get further guidance and help through consultation with a SLP.

1. The patient shall be able to contact SLPs who are registered on the system for consultation through text or video call.
2. The patient and SLP shall be able to connect with each other through sending a connection request. This will allow the SLP and patient to contact each other.
3. The SLP shall be able to view the feedback reports of the patient. This will allow the SLP to better guide the patient with speech therapy advice.
4. The SLP shall be able to view the details of the patient as well.

1.3.6 Information Centre

This module is concerned with educating and informing the patients about Speech Disfluencies and Speech Therapy.

1. The system shall be able to provide information related to speech therapy and speech disfluencies to the patient.
2. The system (chatbot) will be able to answer the queries of the patient related to Speech Disfluencies and Speech Therapies. The chatbot will not answer any other queries.

1.4 User Classes and Characteristics

Identify the various user classes that you anticipate will use this product, and describe their pertinent characteristics.

Table 1.1: User Classes

User class	Description
Person Who Stutters (PWS)	A person who stutters is someone who suffers from speech disfluencies. This is the primary end-user of the system. Uses the application in order to improve the fluency of their speech and better understand their speech disfluencies through AI detection, classification and speech analysis. They use the speech therapy exercises on the system to improve their speech fluency.
Speech Language Pathologist (SLP)	A speech language pathologist is someone who is trained on providing speech therapy to people who suffer from speech disfluencies. This is the primary end-user of the system who provides guidance and support to the patients through consultation (video or text) and monitors the patients.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case Diagram

The use case diagram clearly shows the actors of the system and all the use cases. It shows external actors as well along with how each actor interacts with the use cases. The following figure is the use case diagram of our entire system:

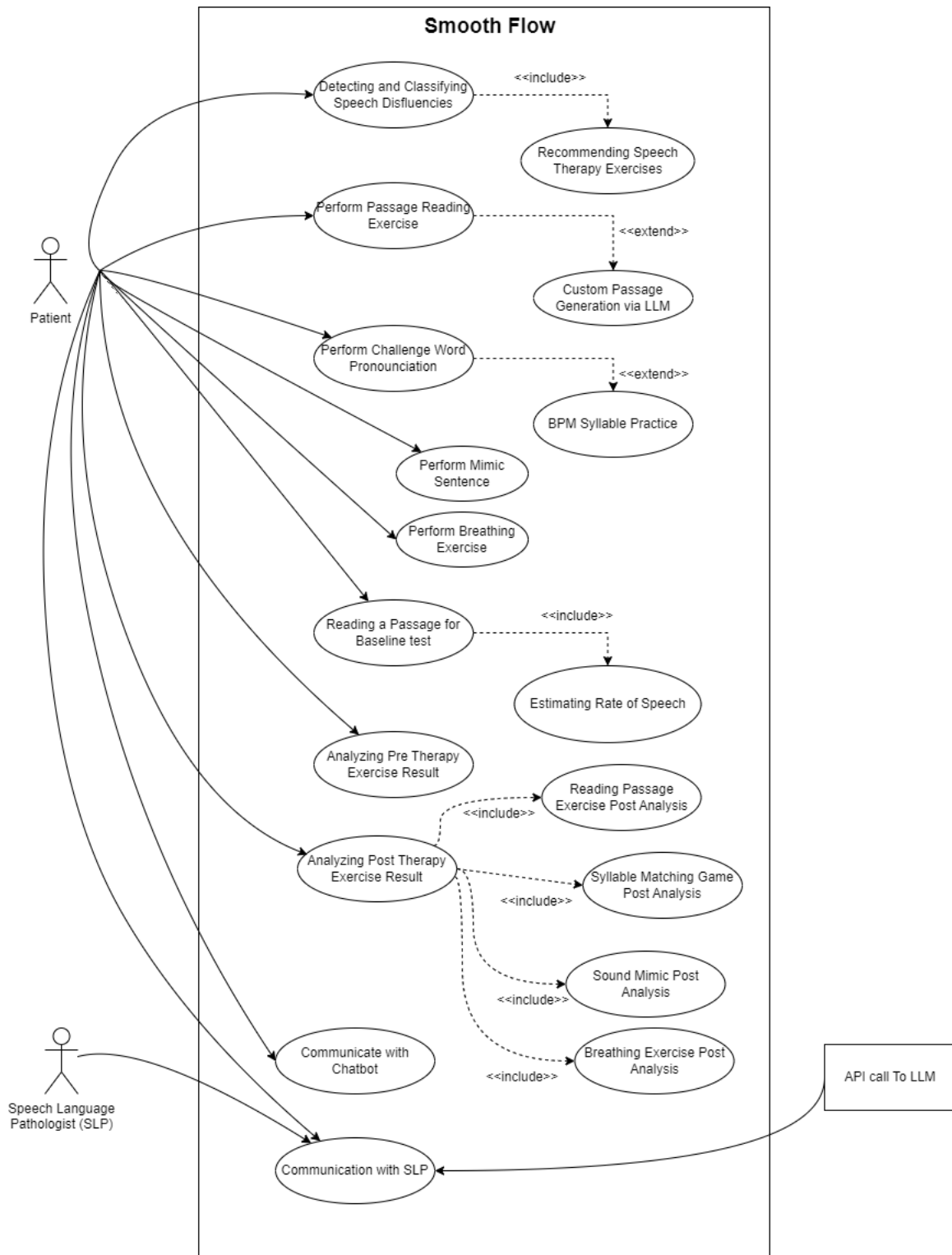


Figure 2.1: Use Case Diagram of Smooth Flow

2.2 Functional Requirements

The following sections contain the Functional Requirements for each module. The functional requirements are described in a high-level use case format.

2.2.1 Module 01: Detection and Multi-label Classification of Patient's Speech

Following are the requirements for module 01:

Use Case 1.0.0	
Use Case Name	Detect and Classify Speech Disfluency
Actors	Patient, System
Type	Primary
Description	The patient records their audio. A sample passage is given to the patient which they can read. The audio gets processed by the system. The system performs multi-label classification on the audio. After processing, the results are provided to the patient.

Use Case 1.1.0	
Use Case Name	Speech Therapy Exercise Recommendation
Actors	System
Type	Primary
Description	After the multi-label classification has been performed on the audio of the patient, the system will recommend speech therapy exercises to the patient based on the classification.

2.2.2 Module 02: Speech Therapy Exercises

Following are the requirements for module 02:

2.2.2.1 Reading Passage Exercise

Use Case 2.0.0	
Use Case Name	Read Passage
Actors	Patient, System
Type	Primary
Description	A passage shall be generated by LLM. The patient shall read the passage while their speech is being recorded by the system. The recorded audio of the patient shall be transcribed into text for further processing.

Use Case 2.0.1	
Use Case Name	Custom Passage Generation via LLM
Actors	Patient, System
Type	Secondary
Description	The system generates a custom passage using LLM based on the patient's selected difficulty. The patient reads the passage, and the system records and transcribes the text for further processing.

2.2.2.2 Matching Syllables Exercise

Use Case 2.1.0	
Use Case Name	Challenging Word Pronunciation
Actors	Patient, System
Type	Primary
Description	The patient selects a challenging word through input. They practice pronouncing the word with spaced-out syllables using a metronome. The patient's challenge word pronunciation will be recorded by the system after the exercise ends for further processing.

Use Case 2.1.1	
Use Case Name	BPM Syllable Practice
Actors	Patient
Type	Secondary
Description	The patient shall be able to adjust the BPM of the metronome before starting the matching syllable exercise. The available BPM options would be 64, 92, 118 and 144.

2.2.2.3 Sound Mimic

Use Case 2.2.0	
Use Case Name	Mimic Pre-recorded Sentence
Actors	Patient
Type	Primary
Description	The patient listens to a pre-recorded sentence and then repeats it. The system records and transcribes their speech for further processing.

2.2.2.4 Breathing Control

Use Case 2.3.0	
Use Case Name	Breathing Animation
Actors	Patient, System
Type	Primary
Description	The system displays a breathing animation. The patient follows visual animations guiding breathing exercises to control anxiety and improve relaxation.

2.2.3 Module 03: Baseline of the Patient

Use Case 3.0.0	
Use Case Name	Passage Reading for Baseline Analysis
Actors	Patient
Type	Primary
Description	The system shall provide a set of passages for the patient to read. These passages start with high complexity and gradually become simpler. The patient reads each passage while the system records their speech for analysis.

Use Case 3.0.1	
Use Case Name	Estimation of Normal Rate of Speech
Actors	System
Type	Primary
Description	Once the system detects that there are no more speech disfluencies, it shall calculate the patient's normal rate of speech. This normal rate will be stored in the patient's profile for use in future comparisons and progress assessments.

2.2.4 Module 04: Progress Tracking and Feedback Reports

Use Case 4.0.0	
Use Case Name	Post Therapy Exercise Analysis
Actors	System, Patient
Type	Primary
Description	After every speech therapy exercise, an analysis would be conducted on the patient's speech to check the disfluency severity.

Use Case 4.1.0	
Use Case Name	Pre Therapy Exercise Analysis
Actors	System, Patient
Type	Primary
Description	Before every speech therapy exercise, an analysis would be conducted on the patient's speech to measure the current state of the patient's speech. The patient would read a passage before completing a speech therapy exercise. During this analysis the audio of the patient would be recorded and then transcribed. The analysis would be conducted on the transcription to determine the disfluency severity.

Use Case 4.2.0	
Use Case Name	Feedback Report Generation
Actors	System
Type	Primary
Description	Based on the post and pre speech therapy exercise analysis done on the patient's speech, the system would generate a feedback report after the speech therapy exercise is completed and the patient has done the post speech therapy exercise test. The feedback report provided by the system will contain the patient's performance (disfluency severity) in the speech therapy exercise, difficulty setting of the exercise, how much the patient improved, time taken to complete the exercise, any achievement badges unlocked and the amount of XP gained.

Use Case 4.3.0	
Use Case Name	Reading Passage Exercise Post Analysis
Actors	System, Patient
Type	Primary
Description	Analysis on the patient's speech will be done after they complete the reading passage exercise. For the analysis, the patient's speech would be recorded by the system as they are reading the passage during the exercise. The recorded speech will be transcribed to text. The transcription would be compared with the original passage. Any mismatch would be counted along with the disfluency severity. The results of the analysis would be provided to the patient through the feedback report.

Use Case 4.4.0	
Use Case Name	Matching Syllables Exercise Post Analysis
Actors	System, Patient
Type	Primary
Description	Analysis on the patient's speech will be done after they complete the matching syllable exercise. For the analysis, the patient's challenge word pronunciation shall be recorded and transcribed to text in order to compare their pronounced syllables to the actual word which shall determine their performance. The results of the analysis would be provided to the patient through the feedback report.

Use Case 4.5.0	
Use Case Name	Sound Mimic Exercise Post Analysis
Actors	System, Patient
Type	Primary
Description	Analysis on the patient's speech will be done after they complete the sound mimic exercise. For the analysis, the patient's speech will be recorded as they are trying to mimic the played recording. The patient's speech will be transcribed and the transcription will be compared with the transcription of the played recording. Any mismatch will be counted along with the disfluency severity. The results of the analysis would be provided to the patient through the feedback report.

Use Case 4.6.0	
Use Case Name	Breathing Exercise Post Analysis
Actors	System, Patient
Type	Primary
Description	Analysis on the patient's speech will be done after they complete the sound mimic exercise. For the analysis, the patient will be provided with a reading passage. As they read, the patient's speech will be recorded. The recorded speech will be transcribed to text. The analysis would be conducted on the transcription to determine the disfluency severity. The results of the analysis would be provided to the patient through the feedback report.

2.2.5 Module 05: SLP Consultation

Use Case 5.0.0	
Use Case Name	Consultation with SLP
Actors	Patient, SLP
Type	Primary
Description	The patient shall be able to contact an SLP who is registered on the system through text message

2.2.6 Module 06: Information Center

Use Case 6.0.0	
Use Case Name	Chatbot Query for Speech Therapy
Actors	Patient, System
Type	Primary
Description	The patient can use the chatbot to ask questions about speech therapy and speech disfluencies. The chatbot will provide answers related to speech therapy and speech disfluencies only.

2.3 Non-Functional Requirements

The non-functional requirements for the project shall primarily follow standards for web based applications. The constraints are as follows:

2.3.1 Reliability

Our application is a web-based application, the system must adhere to standard practices and expectations of web applications. The system shall provide smooth functionality of its features. The system shall have minimal downtime so to make sure user experience is not hindered. The system shall load in efficiently within 10 seconds and constantly provide the patients with relevant exercises and patient data.

REL-1: The system shall have a 99 percent up-time to ensure availability of the system.

REL-2: The system shall have backups to make sure data is not lost in case of failure.

2.3.2 Usability

The system shall be easy to use for patients and SLPs for their respective needs. The system makes sure that patients can focus on improving their speech through the exercises and be give instructions on the therapy exercises.

USE-1: The system shall provide a clear and intuitive user interface to make it easy for patients to navigate through different features and exercises.

USE-2: The system shall display instructions before each exercise to help understand how to complete tasks.

2.3.3 Performance

The system shall have fast response time during low and peak hours of usage.

PER-1: The system shall have a response time of less than 4 seconds.

PER-2: The system shall process and analyze speech audio within 10 seconds.

PER-3: The system shall be able to support up to 500 users.

2.3.4 Security

As the project is a web-based application, web-based security measures will be mandatory for our system.

SEC-1: The system shall use encryption while storing critical information in the database

SEC-2: The system shall have user authentication and authorization.

SEC-3: The system shall have input validation.

SEC-4: The system shall have role-based access control for administration of the system.

Chapter 3

System Overview

The system has 6 major modules that allow it to function. The patient user class can interact with all the modules as they are one of the primary users of the system. The patients can access the detection and classification of their audio for speech disfluencies, perform speech therapy exercises, account creating and management, view progress and feedback reports, consult with an SLP and use the information center with a chatbot. The SLPs can access the SLP consultation module where they can communicate with patients that wish to interact with them over a 2-way communication text channel.

3.1 Architectural Design

The Architecture of Smooth Flow follows a Hybrid Pipe-and-Filter and Repository architecture where each module of the system is connected to a central database through which they all communicate.

The following image is a diagram that encapsulates all the modules and subsystems within the architecture of the application. The Architecture diagram shows how the user communicates with the modules through an API that controls the request and response between the user and the modules of the system. The modules are connected to repositories for storing information and data that is shared and accessed through the central database. The modules communicate through the central database to retrieve and store relevant information,

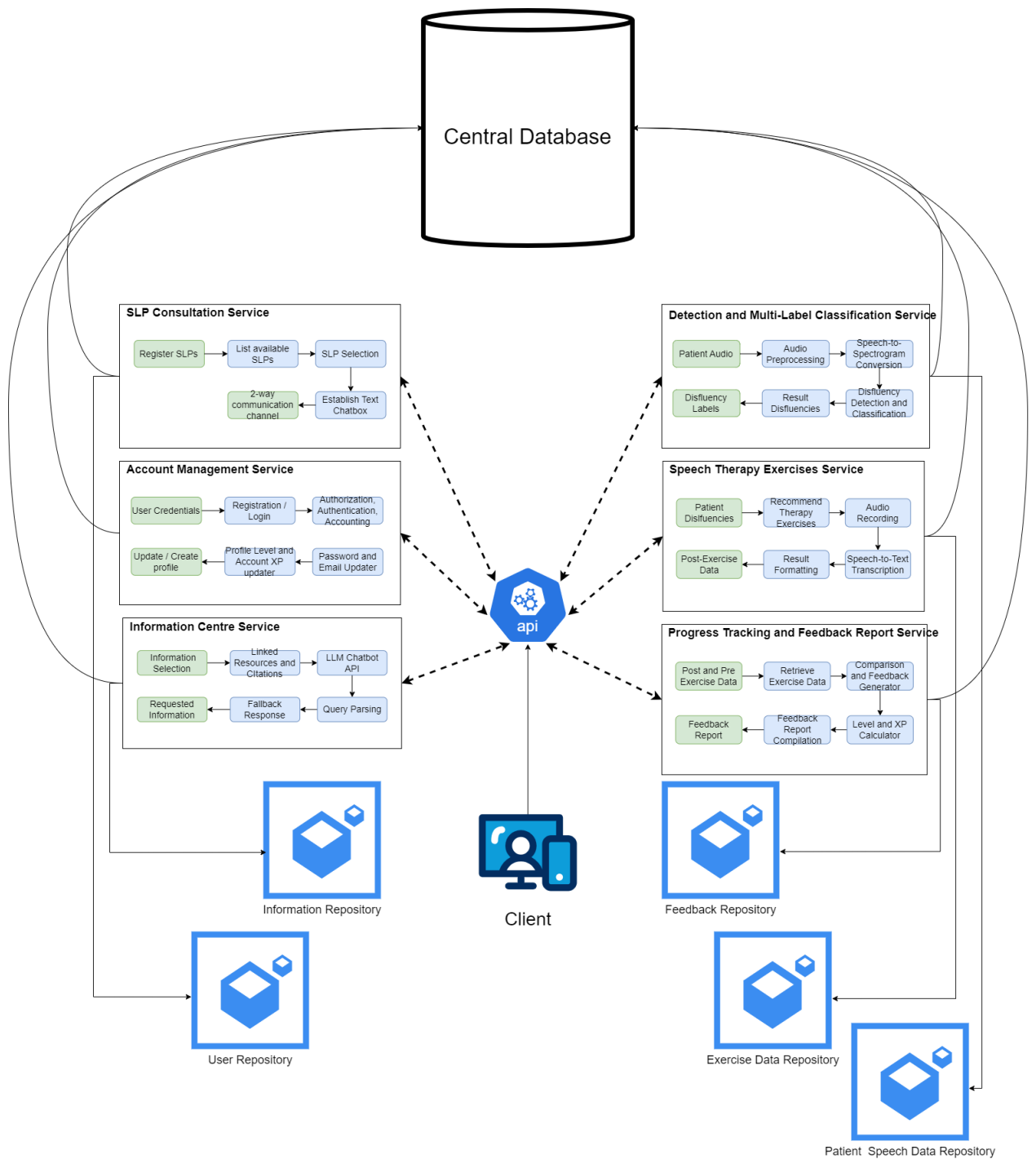


Figure 3.1: Architecture Diagram of Smooth Flow

3.2 Design Models

Design Models for Procedural Approach

The following models show the activity diagrams for each high level use case of the system:

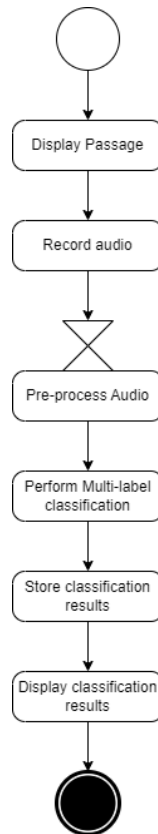


Figure 3.2: Activity Diagram for Use Case 1.0.0 - *Diagram shows the flow of a patient recording their audio. The audio is preprocessed and classified by our fine-tuned model and the labels are displayed to the patient*

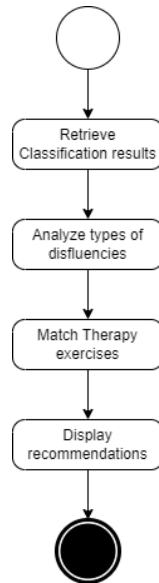


Figure 3.3: Activity Diagram for Use Case 1.1.0 - *Diagram shows how the system will recommend the user speech therapy exercises based on the labels that they get from the fine-tuned model*

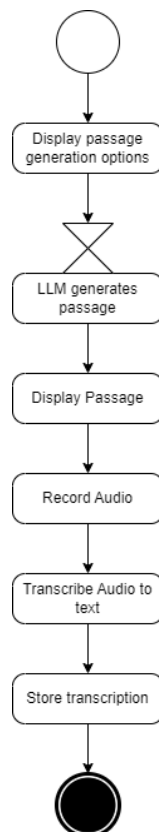


Figure 3.4: Activity Diagram for Use Case 2.0.0 - *Diagram shows the flow of the reading passage exercise, where the user's audio is transcribed after reading the passage,*

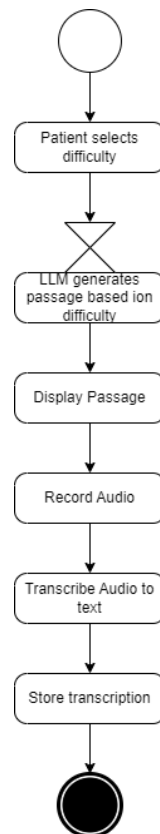


Figure 3.5: Activity Diagram for Use Case 2.0.1 - *Diagram shows how selecting the difficulty affects the content of the passage that is generated to the patient*

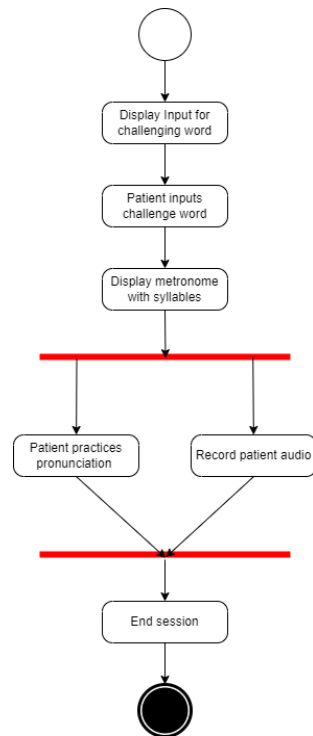


Figure 3.6: Activity Diagram for Use Case 2.1.0 - *Diagram shows the flow of the syllable pronunciation game where the patient selects a challenging word, the system splits into a syllables and their audio is recorded as they attempt to pronounce the word.*

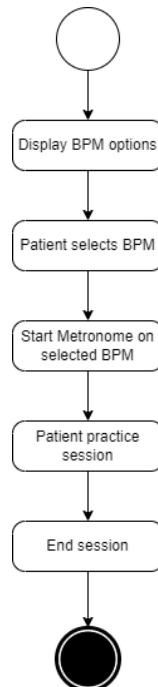


Figure 3.7: Activity Diagram for Use Case 2.1.1 - *Diagram shows how the user can select a different BPM for the speed of the metronome and syllables flashing when helping them pronounce the challenge word*

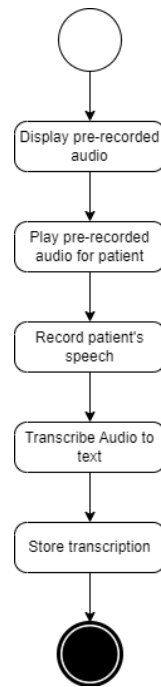


Figure 3.8: Activity Diagram for Use Case 2.2.0 - *Diagram shows the flow of the sound mimic exercise where the patient plays back an audio recording of a properly spoken sentence, which they then have to repeat. The patient's audio is recorded and stored as a transcription,*

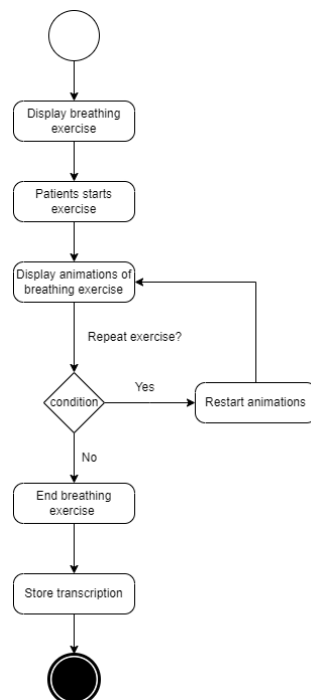


Figure 3.9: Activity Diagram for Use Case 2.3.0 - *Diagram shows the flow of the breathing exercise. The patient follows the breathing exercises and then proceeds to speak a sentence through which their audio is recorded and transcribed.*

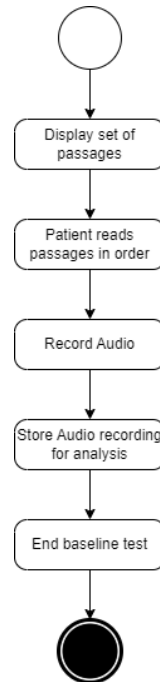


Figure 3.10: Activity Diagram for Use Case 3.0.0 - *Diagram shows the flow of the baseline test that the patient shall attempt to calibrate and calculate their rate of speech and stutter frequency. The patient is provided with several paragraphs with decreasing difficulty.*

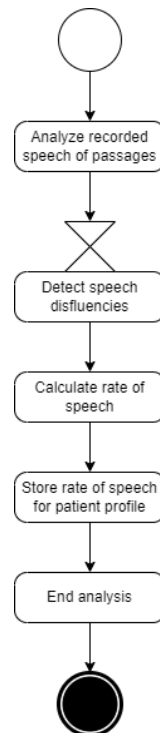


Figure 3.11: Activity Diagram for Use Case 3.0.1 -

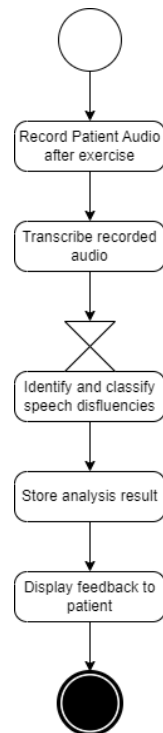


Figure 3.12: Activity Diagram for Use Case 4.0.0 - *Diagram shows the process of the post-exercise analysis of the patient*

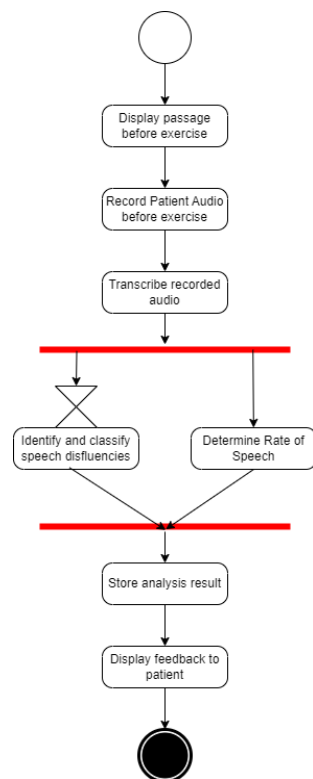


Figure 3.13: Activity Diagram for Use Case 4.1.0 - *Diagram shows the process of the pre-exercise analysis of the patient*

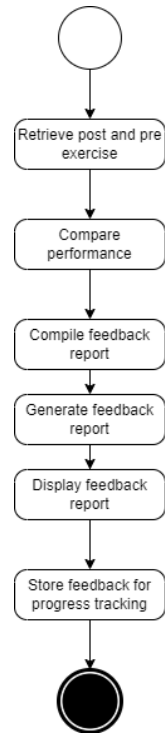


Figure 3.14: Activity Diagram for Use Case 4.2.0 - *Diagram shows the comparison between the patient's pre-exercise and post-exercise data and then generate a feedback report to the patient*

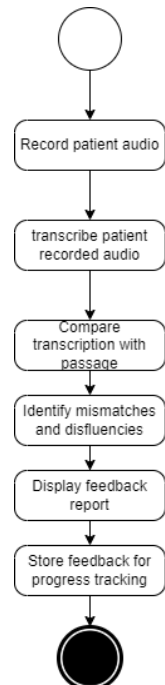


Figure 3.15: Activity Diagram for Use Case 4.3.0 - *Diagram shows the process of analysing the results of the passage reading exercise*

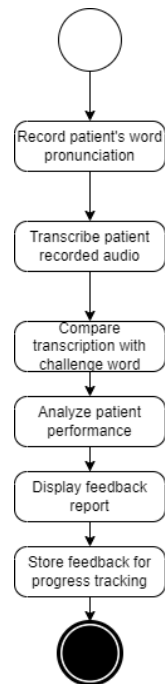


Figure 3.16: Activity Diagram for Use Case 4.4.0 - *Diagram shows the process of analysing the results of the syllable pronunciation exercise*

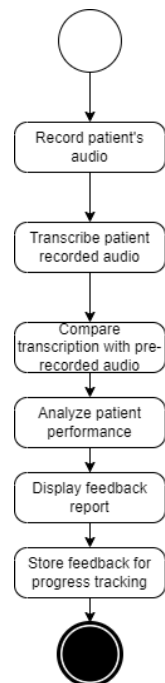


Figure 3.17: Activity Diagram for Use Case 4.5.0 - *Diagram shows the process of analysing the results of the sound mimic exercise*

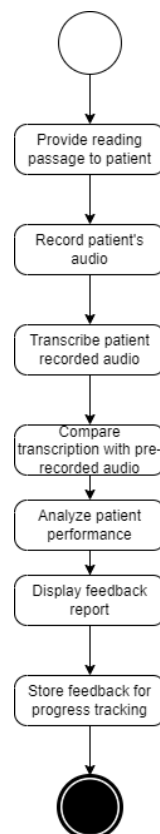


Figure 3.18: Activity Diagram for Use Case 4.6.0 - *Diagram shows the process of analysing the results of the breathing exercise*

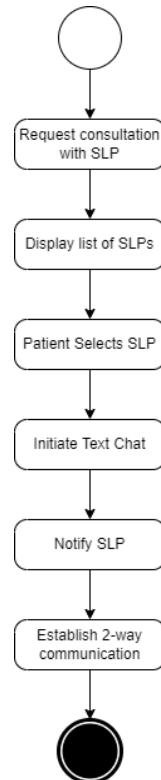


Figure 3.19: Activity Diagram for Use Case 5.0.0 - *Diagram shows the process of how the patient can connect and communicate with an SLP registered in the system*

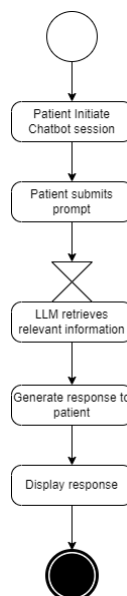


Figure 3.20: Activity Diagram for Use Case 6.0.0 - *Diagram shows the process of how the patient can put in prompts and queries to a chat-bot to learn more about stuttering and speech disfluencies.*

3.3 Data Design

The following figures show the [Data Flow Diagram \(DFD\)](#) of the system. The DFD levels shown start from Level-0 to Level-2. The DFDs have been created representing the entire system's data flow.

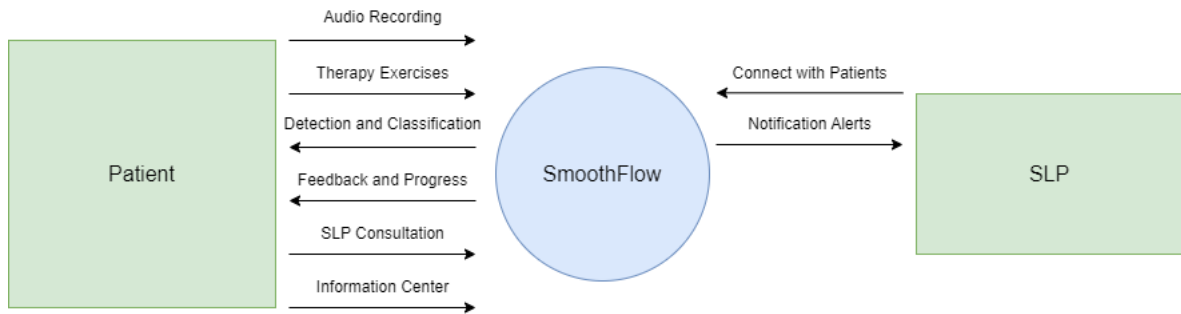


Figure 3.21: DFD Level 0

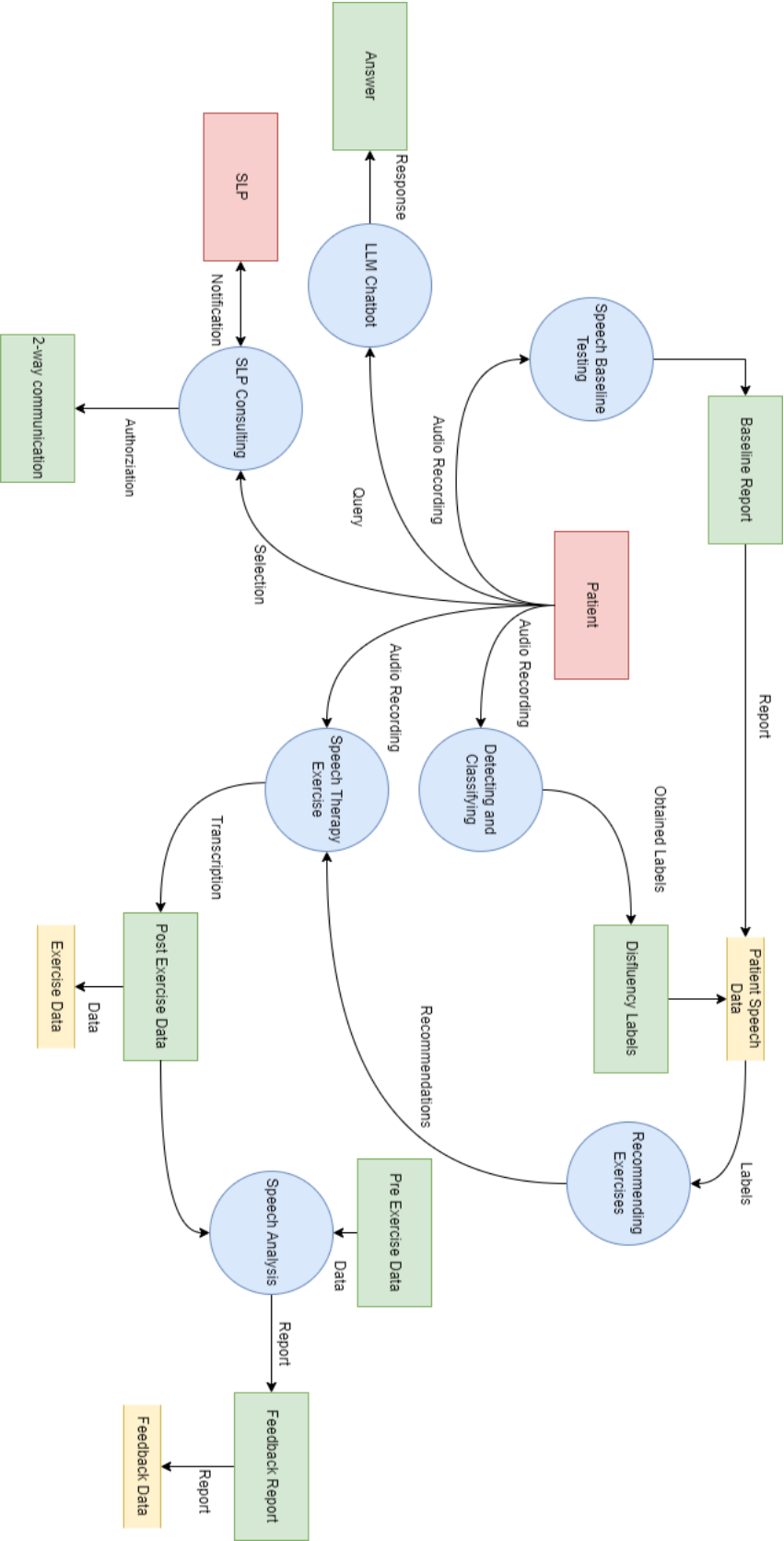


Figure 3.22: DFD Level 1

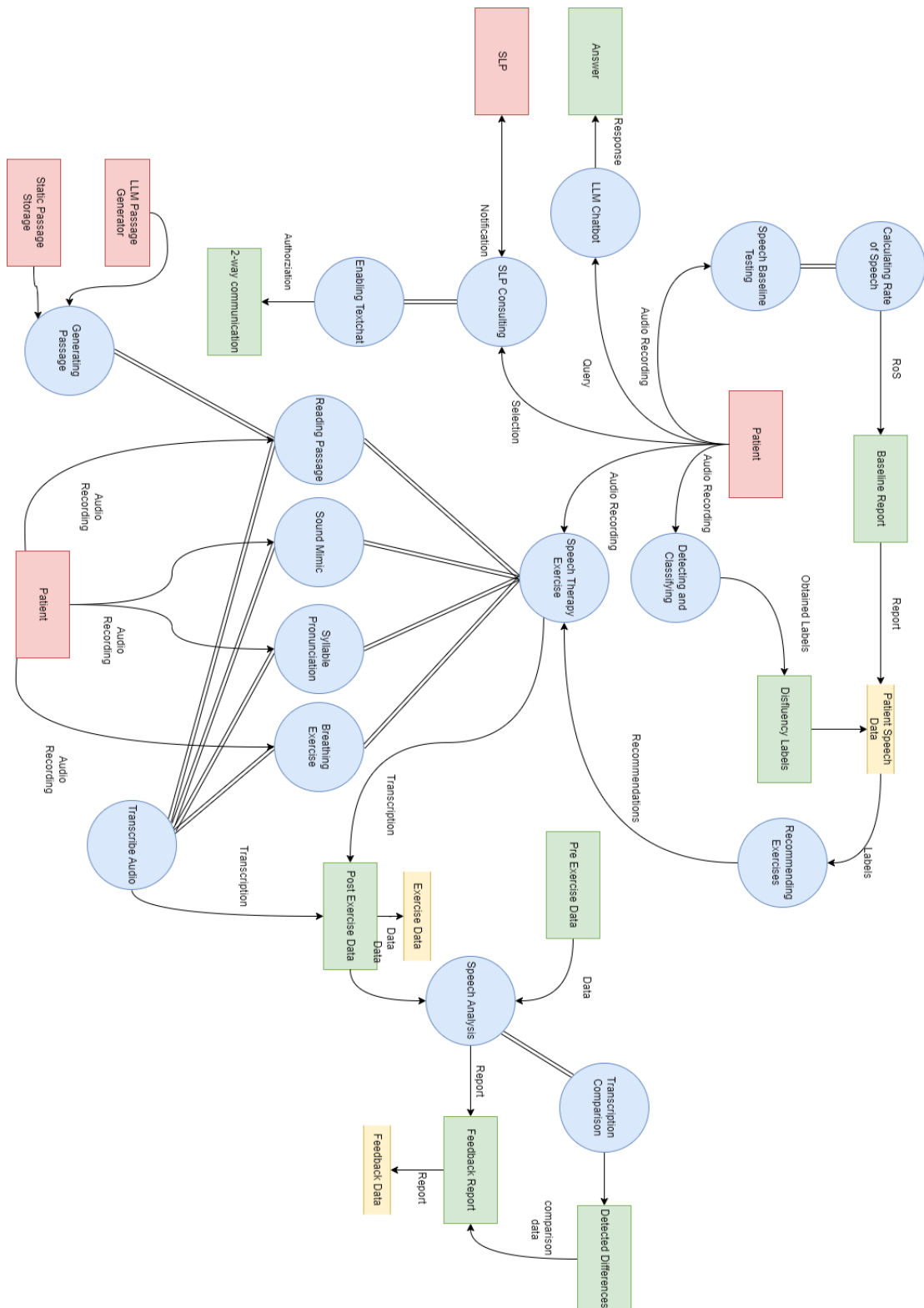


Figure 3.23: DFD Level 2

Appendix A

Dataset

A.1 Title of Dataset

We have utilized two datasets. The title of the first dataset is "Stuttering Events in Podcasts (SEP-28k)" and the title of the second dataset is "Fluency Bank".

A.2 Description of Dataset

Stuttering Events in Podcasts (SEP-28k): A dataset containing over 28k clips (23 hours total) labeled with five event types including blocks, prolongations, sound repetitions, word repetitions, and interjections. Audio comes from public podcasts largely consisting of people who stutter interviewing other people who stutter.

FluencyBank: It is a shared database for the study of fluency development. Participants include typically-developing monolingual and bilingual children, children and adults who stutter (C/AWS) or who clutter (C/AWC), and second language learners. There are around 4k clips in this dataset (3.5 hours total).

A.3 Annotation of Dataset

Audio clips in both datasets were annotated by at least three people who received training via written descriptions, examples, and audio clips on how to best identify each dysfluency but were not clinicians.

The annotation of datasets was done using a variant of time-interval based assessment in which audio recordings are broken into 3 second clips and annotated with binary labels. A clip may contain multiple stuttering event types along with non-dysfluency labels such

as natural pause and unintelligible speech. SEP-28k was also annotated with: no speech, poor audio quality, and music to identify issues specific to this medium

A.4 Objective of Dataset

The main objective of the above mentioned datasets is to train Machine Learning models which can detect and classify the different types of speech disfluencies in the patient's speech.

Appendix B

Gamification Elements for Patient Retention

B.1 Introduction

Gamification refers to "the strategic attempt to enhance systems, services, organizations, and activities by creating similar experiences to those experienced when playing games in order to motivate and engage users".

Speech therapy requires consistency in order to improve the fluency of the speech. In order to encourage patients to stay consistent and better engage them with the speech therapy, we will employ various gamification elements. This will ensure patient retention, better engagement and consistency.

B.2 Description of various Gamification Elements

The different gamification elements that we have planned to utilize in the speech therapy exercises are listed below.

Levelling System: There will be a levelling system for the patients. The patient would be able to level up by gaining the required amount of Experience Points (XP). XP is awarded by playing and completing the Speech Therapy exercises. The amount of XP gained depends on the number of therapy exercises played, percentage improvement and consistency. Each level in the levelling system requires a different amount of XP.

Achievement Badges: The patient will be able to earn Achievement Badges by reaching certain milestones. The patient will be able to display the Achievement Badges on

their profiles. The milestones are:

- Played Speech Therapy Exercise for the First Time
- Played Speech Therapy Exercises daily for X days
- Improved Speech Fluency by 10%
- Achieved Fluency of 80%
- Reached Level 10
- Reached Level 100

Streak: The patient can start a streak by playing speech therapy exercises daily. The streak will be displayed on their profile. The streak would end if the patient misses a day of playing therapy exercises.

Glossary

Block Gasps for air or stuttered pauses. [1](#)

BPM Beats per Minute. [3](#)

DFD Data Flow Diagram. [31](#)

FYP Final Year Project. [1](#)

Gamification the strategic attempt to enhance systems, services, organizations, and activities by creating similar experiences to those experienced when playing games in order to motivate and engage users". [2](#)

LLM Large Language Model. [2](#)

Metronome A metronome is a device that produces an audible click or other sound at a uniform interval. [3](#)

Prolongation Elongated syllable “*M[mmm]ommy*”. [1](#)

PWS People Who Stutter. [6](#)

SLP Speech Language Pathologist. [2](#), [6](#)

XP Experience Points. [1](#)

Bibliography

- [1] ASHA. Definitions of communication disorders and variations. *ASHA Clinical Topics*, 1993.
- [2] Neil R. Carlson. *Physiology of Behavior*. Allyn and Bacon, 11 edition, 2013.
- [3] G. D. Riley. *Stuttering severity instrument*. Pro-Ed, 4 edition, 2009.